

3.3 Risk factors and prevention

Several risk factors for penile cancer have been identified (Table 2) [21] (LE: 2a).

Table 1: Recognised aetiological and epidemiological risk factors for penile cancer

Risk factors	Relevance	Ref
Phimosis	Odds ratio 11-16 vs. no phimosis	[22-24]
Chronic penile inflammation (balanoposthitis related to phimosis), lichen sclerosus	Risk	[25]
Sporalene and ultraviolet A phototherapy for various dermatological conditions such as psoriasis	Incidence rate ratio 9.51 with > 250 treatments	[26]
Smoking	Five-fold increased risk (95% Confidence interval (CI): 2.0-10.1) vs. non-smokers	[22, 23, 27]
HPV infection, condylomata acuminata	22.4% in verrucous squamous cell carcinoma 36-66.3% in basaloid-warty	[13, 28]
Rural areas, low socio-economic status, unmarried		[29-32]
Multiple sexual partners, early age of first intercourse	Three to five-fold increased risk of penile cancer	[21, 23, 33]

Human papilloma virus infection is a risk factor for penile cancer [34]. Human papilloma virus DNA has been identified in 70-100% of intra-epithelial neoplasia and in 30-40% of invasive penile cancer tissue samples (LE: 2a). The HPV virus interacts with oncogenes and tumour suppressor genes (*p16*, *P53*, *Rb* genes) [28, 35]. The rate of HPV-positivity differs between different histological subtypes of penile SCC. Human papilloma virus is a cofactor in the carcinogenesis of some variants of penile SCC, while others are not related to HPV. The commonest HPV subtypes in penile cancer are types 16 and 18 [36]. The risk of penile cancer is increased in patients with condyloma acuminata [37] (LE: 2b).

A significantly better five-year disease-specific survival has been reported for HPV-positive vs. HPV-negative cases (93% vs. 78%) in one study [38], while no difference in lymph node metastases and ten-year survival was reported in another study [39] (Table 3). There is no association between the incidence of penile and cervical cancer, although both are linked to HPV [40, 41]. Female sexual partners of patients with penile cancer do not have an increased incidence of cervical cancer [42].

Table 3: Outcomes for HPV and non-HPV penile carcinomas

Non HPV related	Prognosis	HPV related	Prognosis
SCC usual type/NOS	30% DOD	Basaloid SCC	> 50% DOD
Pseudohyperplastic carcinoma	0%	Papillary basaloid carcinoma	
Pseudoglandular carcinoma	> 50%	Warty carcinoma	Mortality low
Verrucous carcinoma	Good	Warty-basaloid carcinoma	30% DOD
Carcinoma cuniculatum	Good	Clear-cell carcinoma	20%
Papillary carcinoma NOS	Good	Lymphoepithelioma-like carcinoma	Not known
Adenosquamous carcinoma	Good		
Sarcomatoid carcinoma	75% DOD		

DOD = died of disease; HPV = human papillomavirus; SCC = squamous cell carcinoma.

At present, except for a few countries, there is no general recommendation for HPV vaccination in males because of the different HPV-associated risk patterns in penile- and cervical cancer. Furthermore, the epidemiological effects of HPV vaccination in girls still have to be assessed [43, 44].

Phimosis is strongly associated with invasive penile cancer [23, 29, 45, 46], due to associated chronic infection. However, smegma is not a carcinogen [45]. The incidence of lichen sclerosus is relatively high in penile cancer but is not associated with adverse histopathological features, including penile intraepithelial neoplasia (PeIN). Other epidemiological risk factors are cigarette smoking, low socioeconomic status and a low level of education [29, 46].

Neonatal circumcision reduces the incidence of penile cancer; however, it does not seem to reduce the risk of PeIN [23]. The lowest incidence of penile cancer is reported for Israeli Jews (0.3/100,000/year). One matched-pair, case-control study reported that the protective effect of neonatal circumcision against invasive penile cancer (OR 0.41) was much weaker when the analysis was restricted to men without a history of phimosis (OR 0.79, 95% CI: 0.29-2) [23]. Circumcision in adult life does not have any protective effect.

The controversial discussion about neonatal circumcision should take into account that circumcision removes approximately half the tissue that can develop into penile cancer.

3.4 Pathology

Squamous cell carcinoma accounts for over 95% of penile malignancies (see Table 1). It is not known how often SCC is preceded by premalignant lesions (see Table 4) [47-50].

Different histological types of penile SCC with different growth patterns, clinical aggressiveness and HPV associations have been identified (see Table 5). Numerous mixed forms exist such as the warty-basaloid form, with 50-60% the most common mixed form, the usual- verrucous (hybrid), usual-warty, usual-basaloid and the usual-papillary, as well as other rarer combinations.

Other malignant lesions of the penis, all much less common than penile SCC, are melanocytic lesions, mesenchymal tumours, lymphomas and metastases. Penile metastases are frequently of prostatic or colorectal origin. Different types of penile sarcoma have been reported.

Table 4: Premalignant penile lesions (precursor lesions)

Lesions sporadically associated with squamous cell carcinoma (SCC) of the penis:
• Bowenoid papulosis of the penis (HPV related)
• Lichen sclerosis
Premalignant lesions (up to one-third transform to invasive SCC):
• Penile intraepithelial lesions
• Giant condylomata (Buschke-Löwenstein)
• Bowen's disease
• Paget's disease (intradermal ADK)

Table 5: Classification of intra-epithelial neoplasia (PeIN)

• Non-HPV-related PeIN
• o Differentiated PeIN
• HPV-related PeIN
• o Basaloid PeIN
• Warty PeIN
• Warty-basaloid PeIN
• Other rare patterns of PeIN (pleomorphic, spindle, clear cell, pagetoid)

3.4.1 Gross handling of pathology specimens

Tissue sections determine the accuracy of histological diagnosis. Small lesions should be fully included, bigger lesions should have at least 3-4 blocks. Lymph nodes must be included in their entirety after having been inked, in order to detect metastases. After having been inked, surgical margins have to be completely included [51]. Second-opinion pathology review is highly desirable for this rare tumour entity [52].

3.4.2 Pathology report

The pathology report must include the anatomical site of the primary tumour, the histological type of SCC, grade, perineural invasion, depth of invasion, vascular invasion (venous/lymphatic), irregular growth and front of invasion, urethral invasion, invasion of corpus spongiosum/cavernosum, surgical margins and the p16/HPV status (Table 6) [53-56].

Male Genital Lichen Sclerosus

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Abstract

Male genital lichen sclerosis (MGLSc) is a chronic inflammatory skin disease responsible for male sexual dyspareunia and urological morbidity. An afeared complication is squamous cell carcinoma (SCC) of the penis. The precise etiopathogenesis of MGLSc remains controversial although genetic, autoimmune and infective (such as human papillomavirus (HPV) hepatitis C (HCV), Epstein-Barr virus (EBV) and Borrelia) factors have been implicated: Consideration of all the evidence suggests that chronic exposure of susceptible epithelium to urinary occlusion by the foreskin seems the most likely pathomechanism. The mainstay of treatment is topical ultrapotent corticosteroid therapy. Surgery is indicated for cases unresponsive to topical corticosteroid therapy, phimosis, meatal stenosis, urethral stricture, carcinoma in situ (CIS) and squamous cell carcinoma.

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What was known?

MGLSc is a cause of male sexual dysfunction and urological morbidity and associated with squamous carcinoma of the penis. It has been held to be an autoimmune disease.

Introduction

MGLSc is an acquired, chronic inflammatory cutaneous disease of disputed etiology. It causes acute and chronic balanoposthitis and scarring that can result in significant sexual and urinary dysfunction as well as conferring a risk of squamous carcinoma of the penis.^[1,2]

The exact incidence and prevalence are difficult to ascertain. This could be because it is managed by many specialties (dermatology, genito-urinary medicine, urology and pediatrics), as well as significant under-recognition and hence under-reporting by patients and physicians. The estimated incidence of LSc ranges from 1/300 (0.3%) to 1/1000 (0.1%) in both sexes.^[3] A more recent study reported the incidence of MGLSc of 1.4 patients per 100, 000 visit.^[4] The epidemiology may vary between countries and racial groups because MGLSc is principally a disease of the uncircumcised although it can persist or recur after circumcision. There is a bimodal age incidence with peaks in young boys and in adult men, late in the fourth decade.^[4-9]

MGLSc can be the cause of significant morbidity. The symptomatology is an expression of preputial and urethral dysfunction very often summarized as male

dyspareunia.^[2,6,9] The spectrum of presentation is wide. It may be asymptomatic. Some patients describe spontaneous itching, burning, bleeding, tearing, splitting (of the prepuce and frenulum), or hemorrhagic blisters or urinary problems such as dysuria and narrowing of the urinary stream, or anatomical changes of the genitalia. Most men (55%) have symptoms associated with sexual activity but dyspareunia may be even more common than this statistic due to failure or difficulty in eliciting the symptomatology.^[6] Other presentations are non-retractile foreskin (phimosis), foreskin fixed in retraction (paraphimosis), urethritis, urethral stricture, urinary retention and renal failure. Some cases are diagnosed at the presentation of CIS or frank SCC. The disease and its complications have a significant impact on male sexual health.^[6,10]

Signs may be subtle or florid. The organ should be examined systematically and sedulously, including with a hand lens. The classical genital morphologic manifestations of MGLSc are similar to those features of the disease encountered at extragenital sites, and include atrophic ivory-white patches (leukoderma) or plaques, hypertrophic, slightly scaly patches or plaques with telangiectasia and sparse purpura. A constrictive lichenoid posthitis is commonly seen associated with a fibrotic preputial band causing "hourglass" "waisting" of the penile shaft. Variable asymmetric, sometimes florid, Zoonoid inflammation may be present and distract the clinician from the correct diagnosis. The glans may be etiolated, affected by sclerosis especially parametally and show purpura, ecchymosis or angiokeratomas. Meatal involvement may be subtle with encroachment

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disease might be challenging to diagnose clinically against a background of LSc. A retrospective review by Barbagli *et al.* of 130 patients with MGLSc, revealed 11 men (8.4%) with premalignant or malignant features (7 cases with SCC, 2 cases with verrucous carcinoma, 1 case of SCC associated with verrucous carcinoma and 1 case of erythroplasia of Queyrat).^[73] The time interval between diagnosis of LSc and the development of SCC was 14-30 years. Two forms of penile intraepithelial neoplasia (PeIN) have been described (analogy with the vulvar intraepithelial neoplasia classification): Differentiated PeIN which is often associated with MGLSc and undifferentiated or Bowenoid PeIN which is associated with HPV infection.^[74]

The association of GLSc and SCC is widely recognized and the risk has been estimated at between 2% and 12.5%.^[13,73,75-77] MGLSc is also considered one of the two pathways for the development of CIS and penile cancer, the other being via HPV infection.^[2,74] However, in a large series of patients from our clinics not one patient developed SCC, suggesting that accurate diagnosis and definitive curative management are central to arresting carcinogenic drive.

MGLSc and melanocytic lesions

Atypical genital melanocytic naevus has been reported concomitantly in patients with FG- and MGLSc.^[78-80] Cases of melanoma complicating FGLSc have appeared in the literature.^[81-85] We have seen three in MGLSc.^[86] It has been postulated that LSc might be an immune response possibly triggered by the melanocytic lesion, alternatively that the inflammatory environment of LSc provides a microenvironmental niche promoting melanocytic neoplasia.

Diagnosis and Treatment

Most cases of MGLSc can be diagnosed clinically. Under some circumstances a biopsy can be performed. A biopsy is useful to exclude neoplastic transformation and differentiate LSc from erosive lichen planus, scarring cicatricial pemphigoid and lichen simplex.^[1,2] The histology of LSc is characterized by flattening of the epidermis with variable hyperkeratosis, hydropic degeneration of basal cells and an inflammatory cell dermal infiltrate. However, biopsy histology of MGLSc may be non-specific, depending on timing, site and severity.^[87] A biopsy of the most clinically abnormal area, for example, of inflammation or scarring, may not yield a histological diagnosis. There may be nonspecific inflammatory changes (e.g. zoonoid) leading to an erroneous diagnosis of Zoon's balanitis. In many men the preputial or glanular changes may be secondary and reactive to the balanopreputial dysfunction created by the principal proximate or adjacent or apposed site of the LSc. A negative biopsy does not exclude LSc. A negative biopsy does not exclude neoplasia.^[2,9,66,88]

The treatment of MGLSc aims to achieve optimal symptomatic relief, prevention of further deterioration of anatomical architecture changes and a theoretical mitigation of malignant transformation. Our understanding of the role of occlusional contact with urine in the pathogenesis is fundamental to the effective management and the prognosis of MGLSc.^[65,89]

Treatment protocols in GLSc are well-established (level evidence III).^[2,6,66,90] Avoidance of contact with soap and urine are essential. Regular use of skin barrier emollient should be instituted. Short, trimmed pubic hair is recommended to avoid hair trapping between the foreskin and the glans penis and hence minimize irritation, abrasion and consequent inflammation.^[89] Early aggressive treatment may prevent disease progression.^[91] An ultrapotent topical corticosteroid (usually clobetasol propionate (level evidence I)) used under supervision for a finite course is effective^[2,92-94] and usually problem free, but herpes simplex and wart reactivation do occur.^[90,95] prophylactic aciclovir should be prescribed for patients with a history of genital HSV. Most men achieved remission by ultrapotent topical corticosteroid (50-60%).^[6] Topical calcineurin inhibitors should not be used in MGLSc because of a theoretical synergistic risk of SCC.^[6,61,62,69] If maximal conventional medical treatment is not possible or fails then circumcision is recommended. Removal of the foreskin alters the local environment of glans penis: It removes the occlusive and koebnerizing effects of the foreskin and hence mitigates the consequences of post-micturition microincontinence. Circumcision is curatively effective in the majority (at least > 75% but precise long term follow up data are lacking).^[6] Some men might require advanced urethromeatal surgery. Our results show that accurate diagnosis and rational management restores sexual function and mitigates the risk of urinary dysfunction and can conserve the foreskin in about 50%. Furthermore, not one patient of the many hundreds of patients seen in the clinic over 20 years has yet developed penis cancer.^[6]

Conclusion

MGLSc is an acquired, chronic, inflammatory and fibrosing cutaneous disease responsible for significant sexual and urological dysfunction as well as creating a risk of squamous carcinoma of the penis. The etiology of MGLSc is now much better understood. It is not primarily an autoimmune condition. It is likely that an interaction between the irritant effects of urine and other pathogenic factors, such as chronicity, occlusion and as yet undetermined differential epithelial susceptibility, or reaction, to injury, are necessary for the development of MGLSc. Better understanding the pathogenesis of MGLSc is important to minimize sexual morbidity, urological dysfunction, mitigate the cancer risk it generates and preserve the foreskin when possible.